

Supplementary material.

Table S1. Base substitution in the design of pFREM28

Modification	Base position (pFREM28)	Original base	New base
Substitution	624	A	G
Deletion	687	A	-
Substitution	730	G	T
Substitution	731	A	T
Substitution	833	T	G
Substitution	863	C	T
Substitution	872	A	G
Substitution	875	G	A
Substitution	920	G	A
Substitution	977	C	G
Substitution	979	T	G
Substitution	1013	A	G
Substitution	1091	C	G
Substitution	1100	C	G
Substitution	1154	C	G
Substitution	1172	T	C
Substitution	1126	A	G
Substitution	1271	C	G
Substitution	1274	A	G
Substitution	1544	G	A
Substitution	1589	A	G
Deletion	1834	T	-
Deletion	1834	G	-

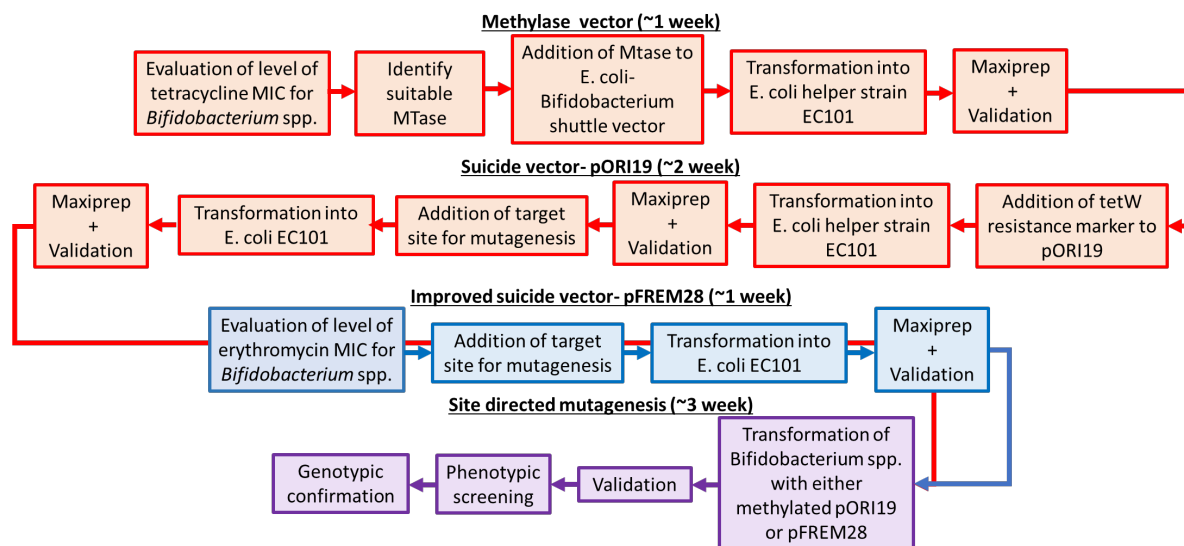


Figure S1. Detailed approaches for site-directed mutagenesis in bifidobacteria. This flow chart describes the original (red pathway ~3 weeks to complete) and modified (blue pathway details specific steps for pFREM28 utilisation, ~1 week to complete) steps required for performing site-directed mutagenesis in *Bifidobacterium* (purple pathway, ~3 weeks work). The improved pFREM28 vector reduces labour and consumable cost as detailed for this workflow when using previous suicide vector pORI19 a step for methylation of the *E. coli-Bifidobacterium* shuttle vector is no longer required.

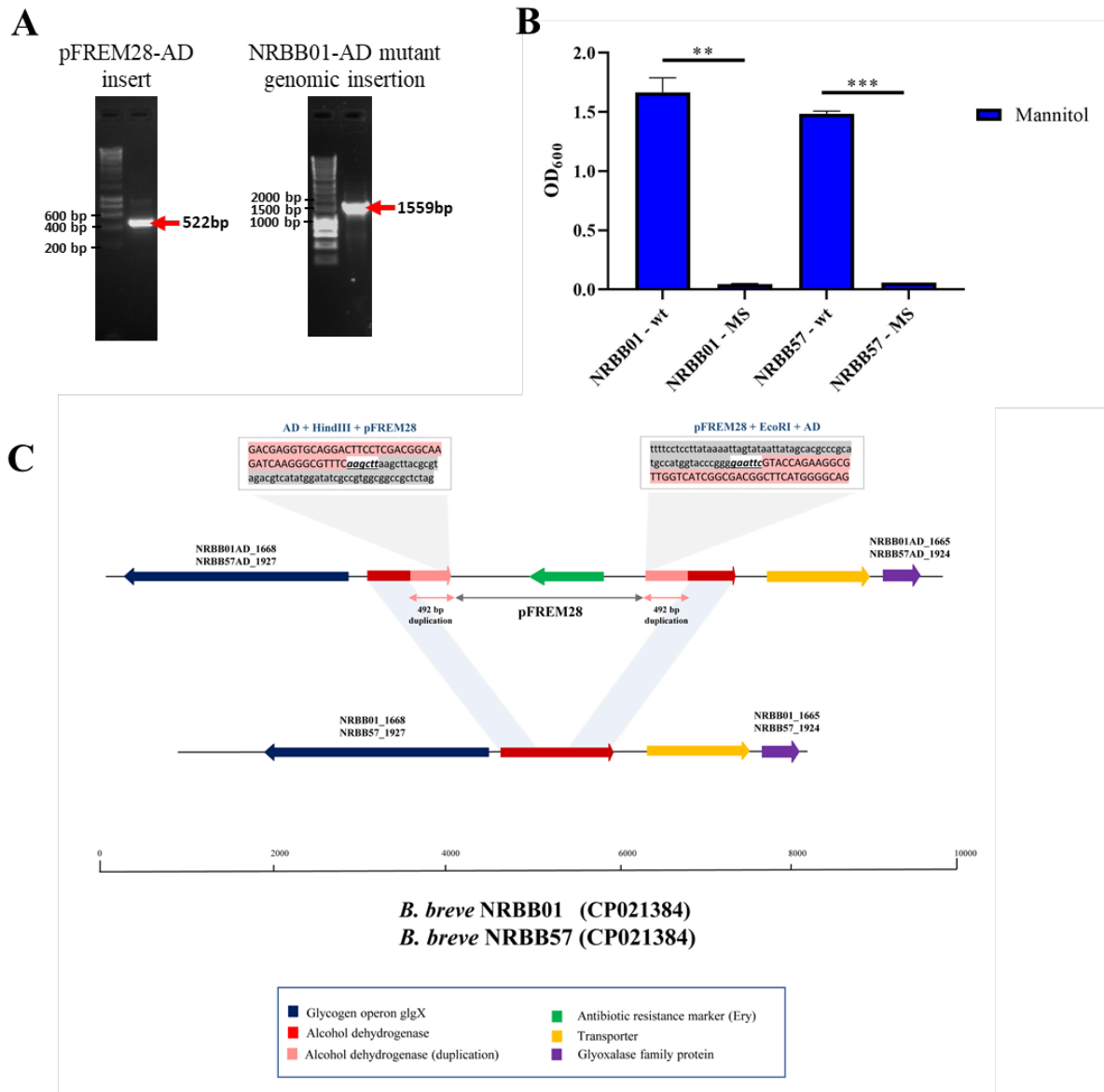


Figure S2. Insertional mutagenesis of the alcohol dehydrogenase-encoding gene in *B. breve*.

A) PCR validation of 492 bp alcohol dehydrogenase-target gene sequence into pFREM28 and subsequent PCR validation of pFREM28-AD insert into the genome of our *B. breve* strains using primers described in Table 3. **B)** Optical density (OD₆₀₀) of *B. breve* strains NRBB01 and NRBB57 wild type and site-directed mutants targeting the alcohol dehydrogenase-encoding gene were grown for 24hrs in the presence of mannitol. Individual values represent the mean ($\pm$ SD) produced from duplicate cultures and significance level (Unpaired t-test): ns, not significant; *, p-value $\leq$ 0.05; **, p-value $\leq$ 0.01; ***, p-value $\leq$ 0.001. **C)** Sequence confirmation of the integration of the pFREM28-AD vector in the alcohol dehydrogenase-

encoding gene of *B. breve* NRBB01 and NRBB57. Locus map showing the comparison between *B. breve* NRBB01 wt harbouring an intact alcohol dehydrogenase-encoding gene (gene ID NRBB01_1667 and NRBB57_1926) and the corresponding locus in the insertional mutant strain containing a disrupted gene. As result of the pFREM28-AD integration (gene ID NRBB01AD_1665 and NRBB01AD_1668; NRBB57AD_1924 and NRBB57AD_1927) the two flanking regions of the integrant present a 492 bp duplication resulting from homologous recombination. All mutants had an erythromycin MIC >256 µg/mL.

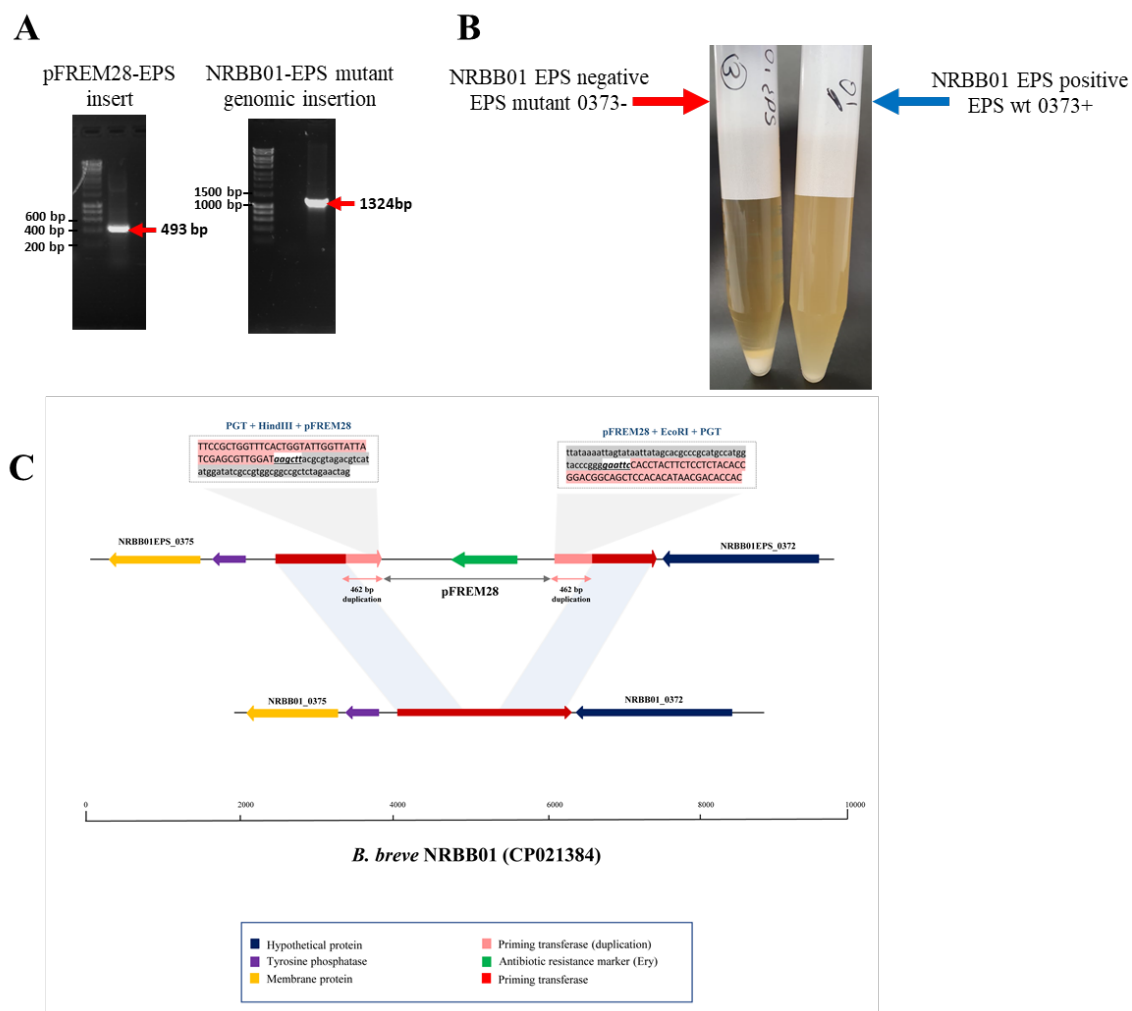


Figure S3. Insertional mutagenesis of the undecaprenyl-phosphate galactosephosphotransferase (Und-GalT)-encoding gene in *B. breve*. **A)** PCR validation of 463 bp Und-GalT-target gene sequence into pFREM28 and subsequent PCR validation of

pFREM28-EPS insert into the genome of our *B. breve* strain using primers described in Table 3. **B)** NRBB01-EPS mutants compared to wild type NRBB01 demonstrate sedimentation from the media due to the interruption of the Und-GalT-encoding gene (gene ID NRBB01_0373). **C)** Sequence confirmation of the integration of the pFREM28-EPS vector in the Und-GalT-encoding gene of *B. breve* NRBB01. Locus map showing the comparison between *B. breve* NRBB01 wt harbouring an intact Und-GalT-encoding gene (gene ID NRBB01_0373) and the corresponding locus in the insertional mutant strain containing a disrupted gene. As result of the pFREM28 integration (gene ID NRBB01EPS_372 and NRBB01EPS_375) the two flanking regions of the integrant present a 463 bp duplication resulting from homologous recombination. All mutants had an erythromycin MIC >256 µg/mL.